

Installing Petalinux 2024.2

- Vitis Unified has already generated platform 'platform_zed_board_linux' with a 'hw' output folder and zed.xsa file (1_Steps.pdf)
- Think we will need to create a 'Linux BSP' using the petalinux tools
- Think Vitis Unified will need some form of the petalinux tool output to run a linux debug session
- This step is seeing if we can successfully install and use the petalinux tools....

We have Ubuntu 24.04.1 LTS running via WSL2 on our Windows 11 PC.

Petalinux Download: <https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/embedded-design-tools/2024-2.html>

Version

[2025.1](#)
2024.2
[2024.1](#)
[Archive](#)

*** Important Information ***

Starting with the 2024.1 release, some PetaLinux BSPs are entering maintenance mode and will no longer be updated. Please consult [this wiki page](#) for details on the AMD support policy for Yocto Project machine configurations.

These pre-built images, source code and configurations are provided for demonstration purposes only and may not be suitable outside of a development environment, including for production purposes.

Prior to production and deployment of any Linux-based system, it is recommended that all relevant security updates are applied, and a mechanism for in-field updates is made available throughout the lifetime of the relevant product.

PetaLinux Tools - Installer - 2024.2 Full Product Installation

Important Information

The PetaLinux Tools installer is downloaded using the below link. The installer checks for the required host machine package requirements followed by license acceptance from the user. It can be installed in any desired path. Note: All BSPs (located below) require the PetaLinux Tools to be installed first.

[PetaLinux Installer \(TAR/GZIP - 2.94 GB\)](#)

MD5 SUM Value : 1c88abb1ba23370692c0aa9d27c0e74

Download Type

Full Product Installation

Last Updated

Nov 18, 2024

Answers

[Release Notes and Known Issues](#)

Documentation

[PetaLinux Tools Documentation](#)
[Installer Information](#)
[What's new in Embedded Software](#)

Downloaded file: petalinux-v2024.2-11062026-installer.run

Copied installer to Ubuntu

In Ubuntu command line:

- Should do these steps before installing petalinux:
 - 73296 - PetaLinux: How to install the required packages for the PetaLinux Build Host:
 - Download 'plnx-env-setup.sh' from 'https://adaptivesupport.amd.com/s/article/73296?language=en_US'
 - Copy script to Ubuntu
 - '[chmod +x plnx-env-setup.sh](#)'
 - '[sudo ./plnx-env-setup.sh](#)'
 - Disable 'dash':
 - Up to Ubuntu 22.04 included (and Debian 11 included):
 - '[sudo dpkg-reconfigure dash](#)' (answer 'No' to transition to bash shell)
 - **Or**, for above version Ubuntu 22.04, try:
 - [sudo ln -s bash /bin/sh.bash](#)
 - [sudo mv /bin/sh.bash /bin/sh](#)
 - And if/when you want to restore back to dash:
 - [sudo ln -s dash /bin/sh.dash](#)
 - [sudo mv /bin/sh.dash /bin/sh](#)
 - Prevention of 'warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)':
 - [sudo apt install locales](#)
 - [sudo locale-gen en_US.UTF-8](#)
 - [sudo dpkg-reconfigure locales](#) (make sure 'en_US.UTF-8' is starred)
 - Prevention of 'package require xsbd FAILED:' problem prevention:
 - [sudo apt update](#)
 - [wget http://security.ubuntu.com/ubuntu/pool/universe/n/ncurses/libtinfo5_6.3-2ubuntu0.1_amd64.deb](http://security.ubuntu.com/ubuntu/pool/universe/n/ncurses/libtinfo5_6.3-2ubuntu0.1_amd64.deb)
 - [sudo apt install ./libtinfo5_6.3-2ubuntu0.1_amd64.deb](#)
- Installing petalinux:

- **'chmod +x petalinux-v2024.2-11062026-installer.run'**
- **'mkdir -p /home/wlaba/playZed/petalinux-tools'**
- **'./petalinux-v2024.2-11062026-installer.run -d /home/wlaba/playZed/petalinux-tools'**

Installer output:

```
./petalinux-v2024.2-11062026-installer.run: line 3: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory
PetaLinux CMD tools installer version 2024.2
```

```
=====
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
[WARNING] This is not a supported OS
[INFO] Checking free disk space
[INFO] Checking installed tools
[INFO] Checking installed development libraries
[INFO] Checking network and other services
[WARNING] No tftp server found - please refer to "UG1144 PetaLinux Tools Documentation Reference Guide" for its impact and solution
```

LICENSE AGREEMENTS

PetaLinux SDK contains software from a number of sources. Please review the following licenses and indicate your acceptance of each to continue.

You do not have to accept the licenses, however if you do not then you may not use PetaLinux SDK.

Use PgUp/PgDn to navigate the license viewer, and press 'q' to close

Press Enter to display the license agreements

```
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Do you accept Xilinx End User License Agreement? [y/N] > y
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Do you accept Third Party End User License Agreement? [y/N] > y
[INFO] Installing PetaLinux SDK...done
[INFO] Setting it up...
```

[INFO] Extracting xsct tarball...

:
:

[INFO] PetaLinux SDK has been successfully set up and is ready to be used.

cd to your installation folder (the folder given after the -d option)

source ./settings.sh

The PetaLinux source code and images provided/generated are for demonstration purposes only.

Please refer to <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/2741928025/Moving+from+PetaLinux+to+Production+Deployment> for more details

PetaLinux environment set to '/home/wlaba/playZed/petalinux-tools'

WARNING: /bin/sh is not bash!

bash is PetaLinux recommended shell. Please set your default shell to bash.

[WARNING] This is not a supported OS

[INFO] Checking free disk space

[INFO] Checking installed tools

[INFO] Checking installed development libraries

[INFO] Checking network and other services

[WARNING] No tftp server found - please refer to "UG1144 2024.2 PetaLinux Tools Documentation Reference Guide" for its impact and solution

Create our first project:

- **cd /home/wlaba/playZed/petalinux-tools** (change directory to our install folder)
- **source ./settings.sh**
- **petalinux-create --type project --template zynq --name PetalinuxDemo1**

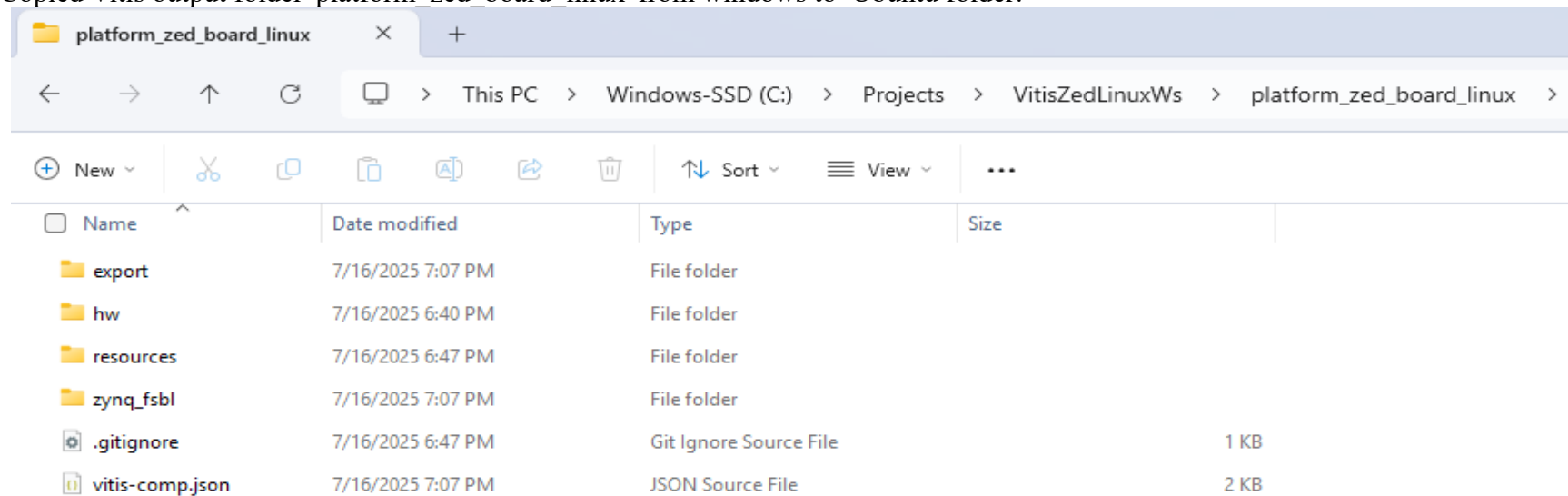
Tool output:

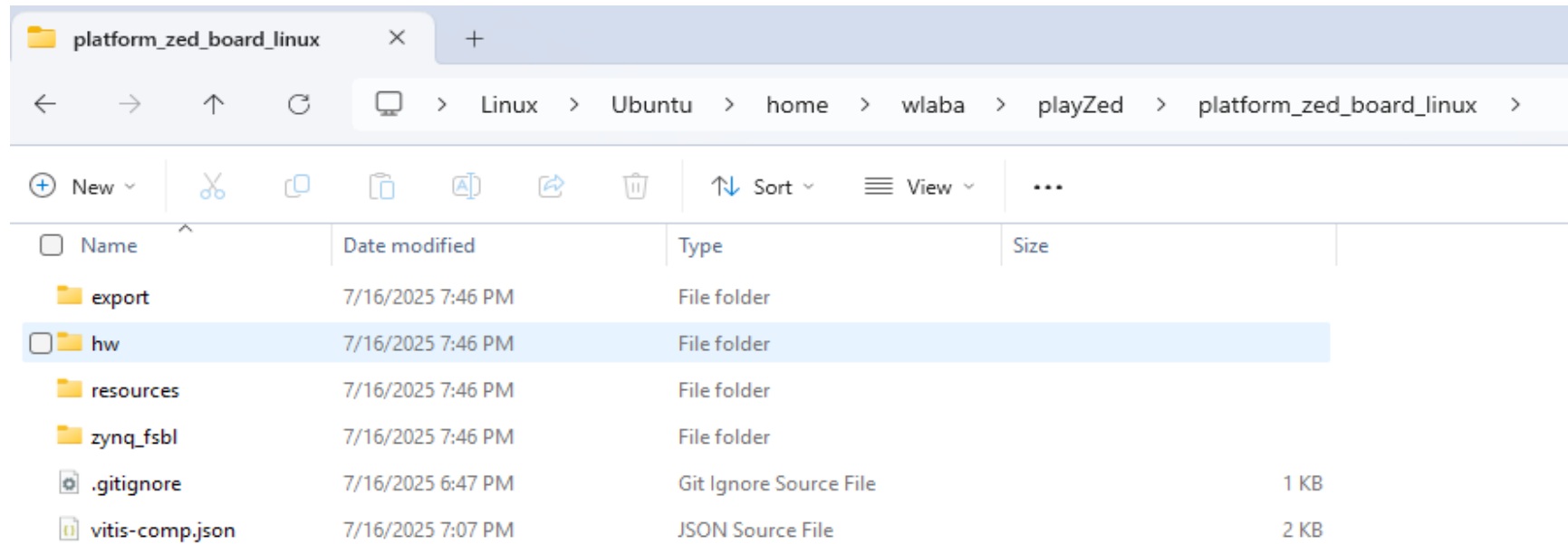
[NOTE] Argument: "--type project" has been deprecated. It is recommended to start using new python command line Argument.

[NOTE] Use: petalinux-create project [OPTIONS]
[INFO] Create project: PetalinuxDemo1
[INFO] New project successfully created in /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1

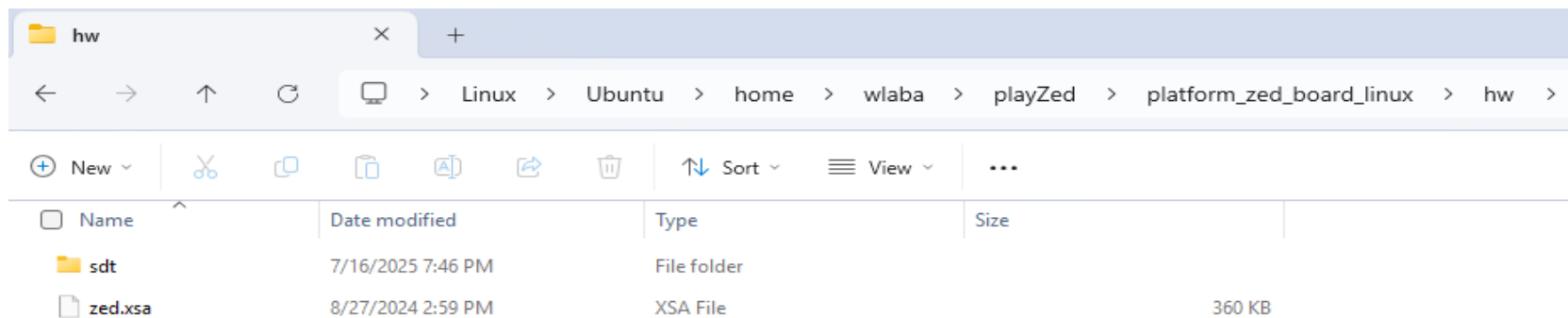
Create our first hardware description output:

- Using 'platform_zed_board_linux' folder generated by Vitis in '1_Steps.pdf'
- Copied Vitis output folder 'platform_zed_board_linux' from windows to Ubuntu folder:





- The 'zed.xsa' file and 'sdt' folder were located under the 'hw' sub-folder:



- `cd /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1` (change directory to our project folder)
- `petalinux-config --get-hw-description=/home/wlaba/playZed/platform_zed_board_linux/hw/`

Tool output:

```
INFO] Getting hardware description
[INFO] Renaming zed.xsa to system.xsa
[DEBUG] gcc --version | sed -ne "s/.* \([0-9]\+\.[0-9]\+\)\.[0-9]\+.*\\1/p"
[DEBUG] unset LD_LIBRARY_PATH;source /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/environment-setup-cortexa9t2hf-neon-xilinx-linux-gnueabi &>> /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/build/config.log;source /home/wlaba/playZed/petalinux-tools/.environment-setup-x86_64-petalinux-linux;source /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/poky/oe-init-build-env /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/build &>> /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/build/config.log;
[DEBUG] unset LD_LIBRARY_PATH;source /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/environment-setup-cortexa9t2hf-neon-xilinx-linux-gnueabi &>> /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/build/config.log;source /home/wlaba/playZed/petalinux-tools/.environment-setup-x86_64-petalinux-linux;source /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/poky/oe-init-build-env /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/build &>> /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/build/config.log;gen-machineconf --soc-family zynq --hw-description /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/hw-description/system.xsa --xsct-tool /home/wlaba/playZed/petalinux-tools/components/xsct --output /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/configs --add-rootfsconfig /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/meta-user/conf/user-rootfsconfig --petalinux --menuconfig project --debug
[INFO] Bitbake is not available, some functionality may be reduced.
[INFO] Using HW file: /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/hw-description/system.xsa
[DEBUG] Using xsct from : /home/wlaba/playZed/petalinux-tools/components/xsct/bin/xsct
[INFO] Getting Platform info from HW file
[DEBUG] Using the soc_family specified by user:zynq
[INFO] Generating Kconfig for project
[DEBUG] Generating System HW file
[DEBUG] xsct -sdx -nodisp /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/gen-machine-scripts/hw-description.tcl plnx_gen_hwsysconf /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/hw-description/system.xsa
Traceback (most recent call last):
  File "/home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/gen-machineconf", line 355, in <module>
    ret = main()
          ^^^^^^
  File "/home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/gen-
```

machineconf", line 349, in main

```
ret = args.func(args)
^^^^^^^^^^^^^^^^^^
```

File "/home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/lib/gen-machineconf/xsct_flow.py", line 325, in ParseXsa

```
hw_info = gatherHWInfo(args)
^^^^^^^^^^^^^^^^^^^^
```

File "/home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/lib/gen-machineconf/xsct_flow.py", line 276, in gatherHWInfo

```
GenXsctSystemHwFile(genmachine_scripts, Kconfig_syshw,
```

File "/home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/lib/gen-machineconf/xsct_flow.py", line 160, in GenXsctSystemHwFile

```
common_utils.RunCmd(cmd, output, shell=True)
```

File "/home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/components/yocto/layers/meta-xilinx/meta-xilinx-core/gen-machine-conf/lib/common_utils.py", line 136, in RunCmd

```
raise Exception("\n%s\n%s\n%s" %
```

Exception:

/home/wlaba/playZed/petalinux-tools/components/xsct/bin/rdiArgs.sh: line 37: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory

/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)

terminate called after throwing an instance of 'std::runtime_error'

```
what(): locale::facet::_S_create_c_locale name not valid
```

/home/wlaba/playZed/petalinux-tools/components/xsct/bin/rdiArgs.sh: line 390: 75274 Aborted

(core dumped) "\$RDI_PROG" "\$@"

[ERROR]

/home/wlaba/playZed/petalinux-tools/components/xsct/bin/rdiArgs.sh: line 37: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory

/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)

terminate called after throwing an instance of 'std::runtime_error'

```
what(): locale::facet::_S_create_c_locale name not valid
```

/home/wlaba/playZed/petalinux-tools/components/xsct/bin/rdiArgs.sh: line 390: 75274 Aborted

(core dumped) "\$RDI_PROG" "\$@"

[DEBUG] Shutting down bitbake

[**ERROR**] Command gen-machineconf --soc-family zynq --hw-description /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/hw-description/system.xsa --xsct-tool /home/wlaba/playZed/petalinux-tools/components/xsct --output /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/configs --add-rootfsconfig /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1/project-spec/meta-user/conf/user-rootfsconfig --petalinux --menuconfig project --debug **failed**

- Looks like petalinux 2024.2 was installed on Ubuntu successfully
- Looks like petalinux project was created successfully
- However, petalinux configure failed!
- More research is needed!....